Abduction as Overwrite: Negative Capability as a Structural Condition for Reliable Inference

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Abstract

The abductive-corrective model of inference treats hypothesis revision as a correction operation within a fixed representational state. We argue this framing is structurally inadequate. Abduction is not correction but *overwrite*: the adoption of a new hypothesis replaces the previous representational state and thereby restructures all subsequent perception, classification, and inference. This has a critical consequence: a premature overwrite, adopted under insufficient evidence, does not merely introduce a local error but establishes a biased interpretive frame that propagates through all downstream cognition. Reliable abduction therefore requires a prior capacity—which we call *negative abductive capability* (NAC)—to suspend hypothesis commitment under insufficient evidence and maintain a provisional state until the evidence threshold is met. We formalize the overwrite transition, the suspension region, and the commitment threshold, and derive implications for cross-species cognition, neuro-symbolic AI design, and the assessment of trustworthy AI systems. This paper is the third in a series: the first [2] formalizes symbol grounding as a topological structure problem; the second [3] establishes reconfigurable hardware as a structural condition for universal sensory grounding; the present paper introduces the dynamic dimension—how representational states transition, and what makes those transitions reliable.

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1 Introduction

Series note. This paper is the third in a series on the structural conditions for reliable cognition. The first paper [2] formalizes symbol grounding as a topological problem: the grounding-induced topology τ_A on a symbol set determines the geometry of cognitive representations, and the Gromov–Hausdorff cognitive distance d_{cog} measures grounding alignment between agents. The second paper [3] establishes a hardware implementation condition: fixed-dataflow architectures impose a positive lower bound on d_{cog} for sensory spaces not embeddable in their computational graph topology, while reconfigurable architectures (CGLA-class) are structurally positioned to minimize this distortion. The present paper introduces the *dynamic* dimension: how does representational structure update over time, and what structural capacity does reliable updating require? The three papers identify three independent sources of d_{cog} increase and the conditions that suppress each. Each paper is self-contained; familiarity with the series is not assumed.

The standard model of abductive inference—inference to the best explanation—treats hypothesis selection as a search operation: given observations, find the hypothesis that best explains them, subject to revision when better evidence arrives [5, 6]. This framing presents revision as correction: an error in the current hypothesis is detected and patched while the underlying representational architecture remains intact.

We argue that this framing is structurally wrong in a precise sense. When an agent adopts a hypothesis, that hypothesis does not merely occupy a slot in a reasoning system that otherwise remains unchanged. It restructures the agent's interpretive frame: subsequent

observations are filtered, classified, and weighted through the lens of the adopted hypothesis. The revision is therefore not a patch but an *overwrite*: the old representational state is replaced by a new one, and the new one determines how all future input is processed.

This distinction matters because it changes the locus of error. In the correction model, a bad hypothesis is a local error, containable and reversible. In the overwrite model, a bad hypothesis is an *initial condition*: it biases all downstream cognition until it is overwritten by a new state—which is itself an overwrite, not a correction. The question is not whether the hypothesis is eventually revised but whether the accumulated distortion before revision is recoverable. In general, it is not.

The practical implication is that reliable abduction cannot be measured solely by the quality of the hypotheses an agent generates. It depends equally on a prior capacity: the ability to *not* generate a hypothesis prematurely. We call this capacity *negative abductive capability* (NAC), by analogy with Keats's negative capability—the ability to remain "in uncertainties, mysteries, doubts, without any irritable reaching after fact and reason" [1]—recast here as a structural cognitive property rather than a poetic temperament. The name is intended to mark a parallel: just as Keats identified the capacity to dwell in unresolved tension as a prerequisite for poetic and intellectual achievement, we identify the capacity to withhold abductive commitment as a structural prerequisite for reliable inference.

Definition 1 (Negative Abductive Capability (NAC))**.** *An agent A possesses negative abductive capability with respect to hypothesis class $\mathcal{H}$ if:*

1. *A maintains a well-defined suspension region $\mathcal{S}$;*
2. *A's commitment threshold θ^* is calibrated to the evidence requirements of $\mathcal{H}$; and*
3. *A can remain in $\mathcal{S}$ for an extended observation window without degradation of suspension capacity.*

Remark 1 (Terminological note on "overwrite")**.** *We use the term* overwrite *in a precise technical sense that differs from its colloquial meaning. In colloquial usage, overwrite implies deletion of the prior content. Here, overwrite denotes a* frame transition*—the installation of a new top-of-stack frame that becomes dominant under the selection rule—while the prior frame remains archived and selectable. More precisely, overwrite = superposition + selection-dominance: the new frame supersedes the prior not by erasure but by winning the selection competition under typical environmental conditions. The term is retained for its theoretical precision in distinguishing categorially distinct hypothesis transitions from within-frame corrections; readers may substitute "frame-dominant transition" if preferred.*

2 The Overwrite Model of Abduction

2.1 Representational States and Transitions

Let $\mathcal{R}$ denote the space of representational states of a cognitive agent A. A representational state $R \in \mathcal{R}$ encodes the agent's current interpretation of its sensory input: the active hypotheses, the assigned categories, the weighted priors over sensory signals.

Definition 2 (Interpretive Frame)**.** *The interpretive frame $F(R)$ induced by state R is the function $F(R) : \mathcal{X} \to \mathcal{Y}$ that maps sensory inputs $x \in \mathcal{X}$ to interpreted outputs $y \in \mathcal{Y}$ under the current representational state.*

The standard corrective model assumes that hypothesis revision is a local operation: $R_t \to R_{t+1}$ modifies a specific component of R_t while leaving the overall frame $F(R)$ approximately intact. We deny this assumption.

Definition 3 (Overwrite Transition)**.** *An overwrite transition is a state change $R_t \to R_{t+1}$ such that the interpretive frame changes globally: $F(R_{t+1}) \not\approx F(R_t)$. Specifically, the mapping of future sensory inputs $x \in \mathcal{X}$ to interpreted outputs differs systematically between the two frames.*

We distinguish overwrite transitions from mere corrections via the topological structure of the induced partition.

Definition 4 (Categorical Distinctness)**.** *Two hypotheses h and h' are* categorially distinct *with respect to sensory input space $\mathcal{X}$ if the partitions of $\mathcal{X}$ they induce are topologically inequivalent: that is, if there exists an open set $U \in \tau_{h'}$ (the topology on $\mathcal{X}$ induced by h') such that $U \notin \tau_h$. Equivalently, h and h' are categorially distinct if the grounding-induced topology changes: $\tau_{h'} \neq \tau_h$. By contrast, a* correction *is a state change $R_t \to R_{t+1}$ in which the induced topologies are homeomorphic—$\tau_{h_{t+1}} \cong \tau_{h_t}$—so that the partition structure of $\mathcal{X}$ is preserved and only the weighting of existing categories is updated.*

Proposition 1 (Abductive Transitions are Overwrites)**.** *Let A adopt a new explanatory hypothesis h_{t+1} at time $t+1$ that supersedes h_t. If h_{t+1} is categorially distinct from h_t—that is, if h_{t+1} induces a different partition of the sensory input space—then the resulting state transition is an overwrite, not a correction.*

Remark 2. *Proposition 1 does not require that the new hypothesis be wrong. Even a correct overwrite restructures the interpretive frame. The concern is with* premature *overwrite—adoption before sufficient evidence—because the resulting frame distortion persists until the next overwrite, which may be long delayed.*

2.2 Bias Propagation after Premature Overwrite

Definition 5 (Premature Overwrite)**.** *An overwrite transition $R_t \to R_{t+1}$ is premature if the evidence base E_t at time t is insufficient to distinguish h_{t+1} from competing hypotheses $\{h^i\}$ at an acceptable confidence level θ:*

$$P(h_{t+1} \mid E_t) < \theta \quad \textit{and} \quad \exists\, h^i \neq h_{t+1} \textit{ s.t. } P(h^i \mid E_t) \geq \theta.$$

Proposition 2 (Bias Propagation)**.** *Let R_{t+1} result from a premature overwrite at time t, where the adopted hypothesis h_{t+1} is categorially distinct (Definition 4) from the ground-truth hypothesis h^*. Then for all subsequent inputs $x_{t'}$, $t' > t$, the interpreted output $F(R_{t+1})(x_{t'})$ carries systematic bias toward h_{t+1} relative to the ground-truth interpretation under full evidence. Within the committed frame $F(R_{t+1})$, this bias is not reducible by further Bayesian*

updating in the setting of the toy model of Appendix A (binary hypotheses, fixed decision boundary); it can only be eliminated by a subsequent overwrite that instantiates a categorially distinct hypothesis. The general condition under which non-reducibility holds beyond this setting is listed as OP4.

Remark 3 (Scope of non-reducibility)**.** *The non-reducibility claim of Proposition 2 is restricted to the case of categorical distinctness. In settings where h_{t+1} and h^* are* not *categorially distinct—that is, where the induced partitions of $\mathcal{X}$ are homeomorphic and only parameter values differ—standard Bayesian updating* will *converge to the correct posterior with sufficient evidence, and the premature commitment constitutes a recoverable inefficiency rather than a structural artifact. The present claim applies specifically to the case in which the adopted hypothesis restructures the topology of the interpretive frame, not merely its parameter values. This scoping is consistent with the toy model of Appendix A, where the two hypotheses induce different partitions of $\mathbb{R}^2$ and within-frame updating cannot recover the correct partition.*

Remark 4 (Epistemic status of Propositions 1 and 2)**.** *The propositions above are structural-model claims, not formally derived theorems. They follow from the definitions of overwrite transition and interpretive frame under the assumption that the frame $F(R)$ is sufficiently sensitive to R that categorially distinct hypotheses induce distinct partitions of $\mathcal{X}$. This assumption holds in standard probabilistic and Bayesian models of inference, where the prior over hypotheses restructures the likelihood function over inputs. A formal proof in a specific probabilistic setting (e.g., Bayesian update with hypothesis-conditional observation models) is given in Appendix A as a toy model. The propositions as stated are intended as theoretical hypotheses supported by structural argument and the phenomenological evidence reviewed in Section 4; their formal generalization is listed as OP4.*

Corollary 1. *A premature overwrite is not a recoverable inefficiency. It is a structural initial condition that biases downstream cognition until overwritten—at which point the cycle may repeat.*

3 Negative Abductive Capability

3.1 Definition and Formal Characterization

Definition 6 (Suspension Region)**.** *The suspension region $\mathcal{S} \subset \mathcal{R}$ is the set of representational states in which the agent maintains active hypotheses tentatively without committing to any single overwrite. An agent in a suspension state continues to accumulate evidence and update probability weights over competing hypotheses without triggering a frame-restructuring transition.*

Definition 7 (Commitment Threshold)**.** *The commitment threshold $\theta^* \in (0, 1)$ is the minimum posterior probability $P(h \mid E)$ required to trigger an overwrite transition $R_t \to R_{t+1}$ instantiating hypothesis h. An agent with higher θ^* requires stronger evidence before committing; an agent with $\theta^* \to 0$ commits on minimal evidence.*

Remark 5. *NAC is not the absence of abductive capacity. It is an additional, structurally prior capacity that regulates when abductive overwrite is triggered. An agent with high abductive generation capacity but low NAC will produce many hypotheses and commit to them rapidly, accumulating bias. An agent with high NAC can generate hypotheses, hold them in suspension, and commit only when evidence warrants.*

3.2 NAC as a Structural Prerequisite

Proposition 3 (NAC Prerequisite for Reliable Abduction)**.** *Let A be an agent with abductive generation capacity α and negative abductive capability ν. The expected bias accumulated per abductive cycle is monotonically decreasing in ν and independent of α. Abductive reliability therefore depends on ν as a structural prerequisite, not on α alone.*

Central thesis. *Abduction requires negative capability. The power to infer depends on the power to suspend inference.*

4 Evidence from Comparative Cognition

4.1 *C. elegans*: Minimum Implementation of NAC

The predict-and-correct chemotaxis loop of *C. elegans* [8] constitutes the minimum known biological implementation of NAC. The 302-neuron architecture does not commit to a locomotor direction on the basis of a single chemosensory reading; it runs multiple correction cycles—maintaining suspension—before a direction is adopted.

This is not a computational limitation. It is a design constraint: premature directional commitment in a noisy chemical gradient environment would produce systematic locomotor error. The suspension region is implemented in neural dynamics; the commitment threshold is set by the sensitivity of the gradient detection circuit.

OP4 of [2] asks for the minimum neuron count required to implement abductive inference with k hypotheses and m correction steps. We reframe this as: what is the minimum network size to implement a suspension region of depth m for a hypothesis class of size k?

4.2 Honeybee Waggle Dance: Grounded Overwrite

The waggle dance [9] encodes food-source location through bodily kinematics calibrated to flight energy. When a scout returns with new location information, the dance parameters are overwritten—not corrected—by the new data. The overwrite is grounded in the bee's own sensorimotor experience of the flight, which provides the evidence base for commitment.

The quality of the overwrite depends on the quality of the flight experience. A bee that has not completed a full flight to the target has insufficient evidence for a committed dance; the suspension capacity is implemented behaviorally as the requirement to complete the journey.

4.3 Multimodal AI: Missing NAC

Current multimodal AI systems lack a well-defined suspension region. Training procedures commit to grounding assignments—instantiated as weight updates—on the basis of each training batch, regardless of whether the evidence in that batch is sufficient to determine the correct interpretive frame for the target concept. The result is premature overwrite at scale: the model's interpretive frame is structured by training distribution artifacts rather than by the target sensory topology.

Two LLM-specific mechanisms are particularly relevant.

Attention collapse. Transformer attention mechanisms weight input tokens by learned similarity scores. When training data is dominated by a particular co-occurrence pattern, attention heads converge to routinely upweighting that pattern—effectively instantiating a premature overwrite of the interpretive frame for the associated concept. Subsequent inputs that do not fit the dominant pattern are downweighted rather than used to revise the frame. The suspension region collapses: the model commits on the basis of partial evidence at training time and cannot maintain genuine uncertainty about the target interpretation at inference time.

Early commitment in fusion layers. Multimodal systems that combine vision, audio, and text via learned fusion layers commit to cross-modal bindings during training. These bindings are fixed in the weights; the model cannot hold two competing cross-modal interpretations in suspension and defer selection to inference time. The commitment threshold θ^* is effectively set by the training loss landscape rather than by the evidence requirements of the target concept. Systems trained with contrastive objectives (e.g., CLIP-style alignment) are particularly susceptible: the contrastive objective rewards confident cross-modal matching rather than calibrated uncertainty, directly suppressing suspension behavior.

The mechanistic interpretability tools of [10] may provide an empirical route to measuring NAC in trained models directly: if internal states corresponding to suspension (maintained competition between hypotheses) can be identified via attribution graphs, the premature commitment rate becomes measurable (OP3).

This connects to the framework of [2]: the grounding-induced topology τ_A of a system without NAC reflects training conditions rather than sensory structure, and the resulting d_{cog} between the system and a biological agent is a structural artifact of premature commitment, not a recoverable inefficiency. Specifically, premature overwrite at training time may *dynamically increase* d_{cog} by locking the agent into a distorted interpretive frame whose topology deviates from the target sensory topology—a dynamic source of cognitive distance distinct from the static sensory mismatch identified in [2] and the hardware constraint identified in [3].

5 Implications

5.1 Trustworthy AI: Beyond Accuracy

A trustworthy AI system is not merely one with high accuracy. It is one that does not prematurely fix its interpretive frame under insufficient evidence. Current evaluation metrics—accuracy, calibration, F1—do not measure NAC directly. We propose the following additional metrics:

1. **Premature commitment rate (PCR)**: the fraction of overwrite transitions triggered below threshold θ^*.
2. **Overwrite reversibility (OR)**: the extent to which a prior overwrite can be undone by subsequent evidence.
3. **Uncertainty retention (UR)**: the duration over which the system can maintain a suspension state without degradation.
4. **Bias propagation factor (BPF)**: the amplification of interpretive bias across successive overwrite cycles.

5.2 Connection to Topological Grounding Framework

Premature overwrite, in the terms of [2], introduces a distortion in the grounding-induced topology τ_A: the topology reflects the structure of the adopted hypothesis rather than the structure of the sensory space. This distortion increases d_{cog} between the agent and an ideally grounded reference system.

We therefore propose that premature overwrite is a *dynamic source of topological distortion*: it increases d_{cog} not through hardware constraint (the concern of [3]) but through commitment dynamics. The three papers in this series then identify three independent sources of d_{cog} increase:

1. Sensory space mismatch (topology paper)
2. Hardware topology mismatch (CGLA paper)
3. Premature abductive overwrite (present paper)

6 Pilot Empirical Evidence

The framework claims that NAC governs the quality of abductive overwrite: systems with low NAC undergo premature commitment, which dynamically increases d_{cog} relative to an ideally grounded reference. We report a pilot study that provides indirect support for this structural claim.

Setup. We applied RSA to three training checkpoints of Pythia-70m [11]: step 0 (random initialization), step 1000 (early training), and step 143000 (reference). Pairwise dissimilarity matrices were computed over $n = 29$ probe concepts at hidden layers $\ell \in \{2, 4, 6\}$, averaged over three prompt templates. Structural alignment was measured as the second-order RSA correlation $\rho_{\text{cog}}(A, B; \ell)$.

Results. ρ_{cog} increases monotonically with training at all layers (L6: step-0 = 0.017; step-1000 = 0.535), indicating progressive alignment toward the reference rather than abrupt replacement. Cross-model replication with GPT-2 [12] shows $\rho_{\text{cog}}(\text{GPT-2}, \text{Pythia-trained}) \approx 0.42$ versus $\rho_{\text{cog}}(\text{GPT-2}, \text{Pythia-step-0}) \approx 0.10$ across all layers.

Interpretation. Under the overwrite model, if training were pure overwrite (low NAC), each batch would overwrite the previous frame completely, producing chaotic relational geometry at intermediate checkpoints. Instead, the monotone increase in ρ_{cog} is consistent with *accumulation of structure* rather than wholesale replacement—supporting the claim that current training procedures implement low-NAC overwrite with residual structure retention, rather than either ideal NAC (suspension until sufficient evidence) or complete deletion.

The cross-model convergence result is consistent with OP4: two architectures trained independently converge to similar τ_A, suggesting that the relational geometry is determined by the training distribution rather than by the architecture—a signature of premature commitment to training-distribution frames. All code is available at `https://github.com/minamominamoto/grounding-without-reality`.

7 Open Problems

The framework developed here is a theoretical proposal rather than a completed formal theory. The central claims—that abduction is overwrite rather than correction, and that reliable overwrite requires negative abductive capability—are advanced as structural hypotheses supported by definitional argument, phenomenological evidence from comparative cognition, and the Bayesian toy model of Appendix A. Their full generality across cognitive architectures, hypothesis classes, and sensory modalities is not established and constitutes the primary research obligation of the program. The open problems below identify the most pressing obligations.

OP1. Formal characterization of suspension region. Define the suspension region $\mathcal{S} \subset \mathcal{R}$ for specific cognitive architectures (transformer, neuromorphic, biological). What are the minimal architectural requirements for a well-defined suspension region?

OP2. Calibration of commitment threshold. For a given hypothesis class $\mathcal{H}$ and evidence structure, what is the optimal θ^*? Is there a bias-variance tradeoff between premature commitment and excessive suspension?

OP3. NAC measurement. Develop empirical protocols for measuring NAC in AI systems and biological agents. A conceptual proof-of-concept protocol is the following.

For *biological agents*: present ambiguous stimuli (e.g., Necker cube, ambiguous chemical gradients for *C. elegans*) under varying time pressure; measure the latency to behavioral commitment and the reversibility of that commitment upon contradictory evidence. Commitment latency and reversibility operationalize θ^* and the depth of $\mathcal{S}$ respectively. For *AI systems*: Anthropic's circuit tracing tools [10] provide a route to measuring suspension states directly. If attribution graphs reveal internal states in which multiple competing hypotheses are simultaneously active (i.e., no single hypothesis suppresses the others), this constitutes a mechanistic correlate of $\mathcal{S}$. The time from stimulus presentation to suppression of competing features operationalizes commitment latency; reinstatement of suppressed features upon contradictory input operationalizes reversibility. Full formal operationalization is deferred to future work.

OP4. Relationship between NAC and d_{cog}. Establish a formal relationship between the premature commitment rate of a system and the resulting increase in d_{cog} relative to an ideally grounded reference.

OP5. NAC in large language models. Do current LLMs exhibit suspension behavior? Mechanistic interpretability tools [10] may reveal whether internal states corresponding to suspension exist in trained models.

8 Conclusion

Abduction is not correction. It is overwrite. A hypothesis adopted by an abductive system restructures the interpretive frame through which all subsequent input is processed. A premature overwrite therefore does not introduce a local error; it establishes a biased initial condition for all downstream cognition.

Reliable abduction therefore requires a structurally prior capacity: the ability to suspend commitment, maintain a provisional state, and defer overwrite until the evidence threshold is met. This is negative abductive capability.

An agent—biological or artificial—that lacks NAC is not merely a poor abductor. It is a bias amplification mechanism: each premature overwrite biases the next observation, which may trigger another premature overwrite, compounding the distortion.

Trustworthy AI is not AI that always answers. It is AI that knows when not to commit.

Abduction requires negative capability. The power to infer depends on the power to suspend inference.

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A Toy Model: Bayesian Overwrite with Bias Propagation

We provide a minimal formal instantiation of Propositions 1 and 2 in a two-hypothesis Bayesian setting.

Setup. Let $\mathcal{H} = \{h_0, h_1\}$ be a binary hypothesis class. Let the sensory input space be $\mathcal{X} = \mathbb{R}^2$, with two regions: $\mathcal{X}_0$ (associated with h_0) and $\mathcal{X}_1$ (associated with h_1), separated by a decision boundary ∂. An agent receives observations $x_1, x_2, \ldots$ drawn from a mixture distribution $p(x) = \pi_0 p(x \mid h_0) + \pi_1 p(x \mid h_1)$, where $\pi_0 + \pi_1 = 1$.

Suspension state. The agent begins in suspension: $P(h_0 \mid \emptyset) = \pi_0$, $P(h_1 \mid \emptyset) = \pi_1$. After each observation x_t, the agent performs a Bayesian update:

$$P(h_i \mid x_1, \ldots, x_t) \propto P(h_i \mid x_1, \ldots, x_{t-1}) \cdot p(x_t \mid h_i).$$

The agent remains in $\mathcal{S}$ (suspension) as long as $\max_i P(h_i \mid x_1, \ldots, x_t) < \theta^*$.

Overwrite transition. At time t^*, the first time $P(h_j \mid x_1, \ldots, x_{t^*}) \geq \theta^*$ for some j, the agent commits: the interpretive frame is set to $F(R_{t^*}) = p(\cdot \mid h_j)$. Subsequent observations are interpreted under this frame.

Premature overwrite. If $t^* < t^\dagger$, where $t^\dagger$ is the time at which the maximum-likelihood hypothesis becomes stable under the full evidence sequence, the overwrite at t^* is premature.

Bias propagation (formal). After commitment to h_j at t^*, define the bias for subsequent input $x_{t'}$ ($t' > t^*$) as:

$$\beta(x_{t'}) = F(R_{t^*})(x_{t'}) - F(R^*)(x_{t'}),$$

where $F(R^*)$ is the frame induced by the ground-truth hypothesis. For all $x_{t'} \in \mathcal{X}_{1-j}$ (inputs from the non-committed region), $\beta(x_{t'}) \neq 0$ and is not correctable by Bayesian update within the committed frame, since the likelihood under h_j assigns low weight to $\mathcal{X}_{1-j}$ by construction.

Connection to d_{cog}. The grounding-induced topology τ_A of the committed agent partitions $\mathcal{X}$ according to h_j, while the ideal topology τ^* partitions $\mathcal{X}$ according to $h^\dagger$ (the full-evidence hypothesis). When $h_j \neq h^\dagger$ (premature commitment to the wrong hypothesis), the two topologies are not homeomorphic, and:

$$d_{\text{cog}}(A, A^*) = d_{GH}((\Sigma, \tau_A), (\Sigma, \tau^*)) > 0.$$

This is the quantitative expression of the claim that premature overwrite dynamically increases cognitive distance. The magnitude of d_{cog} depends on the extent of topological mismatch between the committed and ideal partitions of $\mathcal{X}$, which in turn depends on the distance of t^* from $t^\dagger$.

NAC as bias suppression. An agent with NAC (θ^* appropriately calibrated) remains in suspension until $t \geq t^\dagger$, at which point the committed frame $F(R_{t^\dagger})$ is homeomorphic to $F(R^*)$, and $d_{\text{cog}}(A, A^*) = 0$. The toy model therefore demonstrates that NAC is not merely a desirable property but a sufficient condition for d_{cog} minimization in this setting.

B Numerical Instantiation of Overwrite as H_1 Destabilization

The bias-propagation claim of Proposition 2 (Section **??**) is formulated structurally: premature overwrite instantiates a categorially distinct hypothesis, which propagates bias through all downstream cognition. This appendix provides a numerical instantiation of that claim in the persistence signature framework of the companion paper [2], operationalizing overwrite as a perturbation of sensory centroids and measuring the resulting distortion of the agent's interpretive-frame H_1 feature.

The experiments establish four structural observations, the last of which takes the form of a closed-form relationship between the cycle geometry of an interpretive frame and the effect of a premature overwrite on that frame's H_1 lifetime.

The setup extends the sixteen-concept construction of the companion paper's Appendix **??**. We focus on the bee agent, whose baseline foraging frame is a six-point persistent H_1 feature with lifetime 0.293.

Overwrite as centroid perturbation

We operationalize the overwrite of a concept c by a continuous parameter $t \in [0, 1]$:

$$G_t(c) = (1 - t)\, G_0(c) + t\, G_{\text{target}},$$

where $G_0(c)$ is c's original sensory centroid and G_{target} is a unit vector in a sensorially distinct subspace representing the incorrect hypothesis that c belongs to a different conceptual category. For the bee agent overwriting NECTAR, G_{target} is placed in the social subspace (channels 3–6), representing the overwrite "NECTAR is a social-communication concept."

The dominant H_1 persistence feature of $(\Sigma, d_{\mathcal{A}_t})$ is measured at each t via the Vietoris–Rips filtration of the angular distance matrix.

Continuous overwrite: graded degradation

Figure 1 (top-left) shows the dominant H_1 lifetime as a function of overwrite fraction t when c = NECTAR. The lifetime decreases monotonically from 0.293 at $t = 0$ to 0.139 at $t \geq 0.65$, a 52.5% reduction. This curve supplies a quantitative form of the "graded distortion" component of Proposition 2: the degree of overwrite determines the degree of structural distortion, with most of the distortion occurring once the overwrite crosses a threshold between partial and substantial commitment.

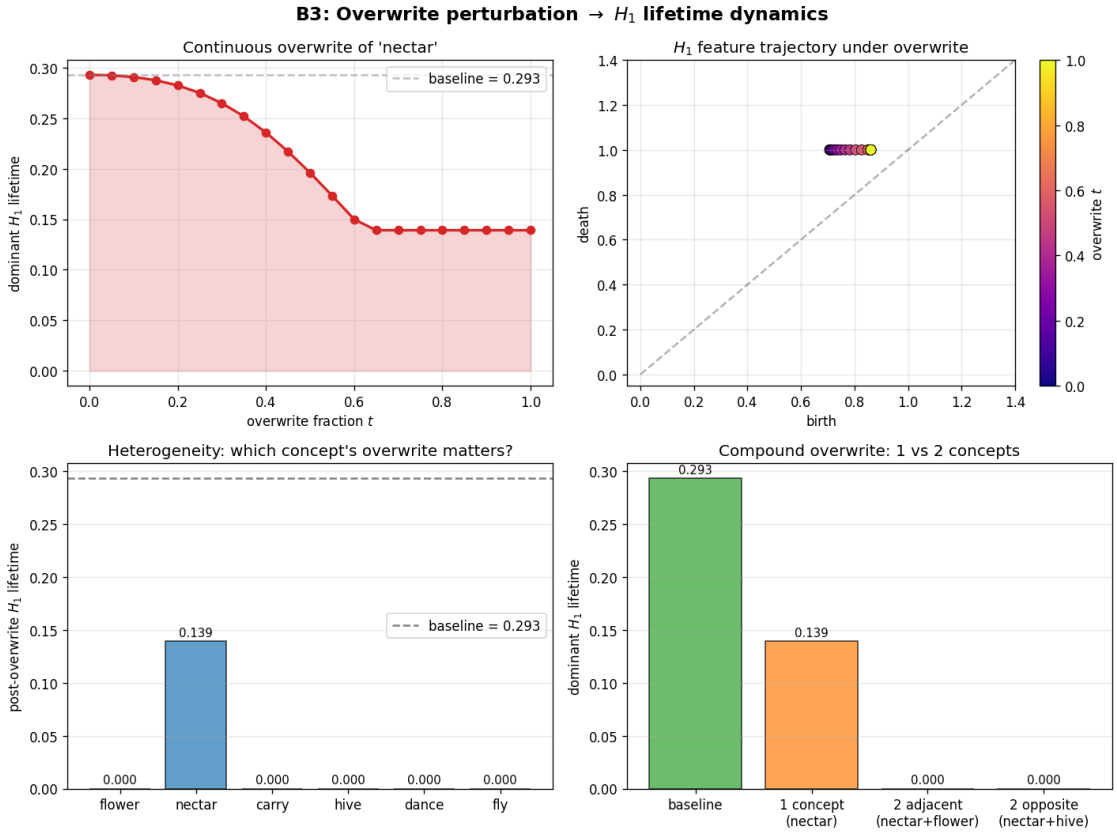

Figure 1: H_1 lifetime dynamics under cross-subspace overwrite perturbation. Top-left: dominant H_1 lifetime as a function of continuous overwrite fraction t for NECTAR. Top-right: trajectory of the H_1 persistence point in the (birth, death) plane. Bottom-left: heterogeneity across loop concepts. Bottom-right: compound overwrite.

Heterogeneity: critical nodes versus structural redundancy

The most theoretically consequential finding of the first set of experiments appears in Figure 1 (bottom-left). Six independent overwrite experiments were performed, each overwriting a single bee-loop concept at $t = 1$ and measuring the residual H_1 lifetime.

Overwritten concept	H_1 lifetime	Reduction
FLOWER	0.000	100%
NECTAR	0.139	52.5%
CARRY	0.000	100%
HIVE	0.000	100%
DANCE	0.000	100%
FLY	0.000	100%

Only NECTAR's overwrite leaves the frame's H_1 feature alive. For the remaining five concepts, their overwrite entirely destroys the feature.

The mechanism is a structural graph-theoretic fact rather than a semantic one. The six bee-loop concepts are arranged on a regular hexagon at angles $\{0, \pi/3, 2\pi/3, \pi, 4\pi/3, 5\pi/3\}$

in the (UV, olfactory) subspace. The non-loop concept FOOD is placed at angle $\pi/4$ in the *same* subspace, near to NECTAR's original angular position of $\pi/3$. When NECTAR is overwritten, the Vietoris–Rips filtration can close a cycle using FOOD as a substitute vertex; the remaining five loop concepts together with FOOD form a six-point cycle with only slightly elevated maximal adjacent gap. No analogous substitute exists in the angular neighborhood of any other loop concept: their removal leaves a $2\pi/3$ gap in the ring, which is wider than the filtration scale at which non-loop concepts cone in, destroying the H_1 feature.

The appropriate terminology for this distinction is graph-theoretic, not semantic. A concept is a *critical node* of an H_1 feature if its removal creates a gap that cannot be bridged by any other concept at a filtration scale below the feature's death; a concept is *redundantly placed* if the surrounding concept landscape contains an alternative vertex that can substitute at a comparable scale. The distinction is a property of the configuration, not of the semantics of the concepts involved; nothing in the analysis depends on what the concepts *represent*, only on how they are positioned. This distinction is purely structural and does not rely on semantic interpretation.

This refines the standard formulation of Proposition 2. The original formulation stated that premature overwrite propagates bias. The refined formulation is: premature overwrite propagates structural distortion to the H_1 feature in proportion to whether the overwritten concept is a critical node of that feature. Critical-node overwrite produces catastrophic feature collapse; redundantly-placed-node overwrite produces graceful degradation with the feature's homology representative migrating to an alternative path through the surrounding landscape. This provides a concrete mechanism by which identical perturbations can lead to qualitatively different downstream effects.

Remark 6 (On terminology)**.** *We deliberately avoid terms such as "semantic centrality" or "conceptual importance" for this distinction. The analysis operates on the angular geometry of centroids in sensory channel space; semantic interpretation of the resulting structural roles is a separate interpretive task. In particular, a concept may be structurally redundant in one agent's sensory topology and structurally critical in another's, precisely because the surrounding concept landscape differs between agents. This relational dependence is the natural topological analogue of the context-sensitivity of frame selection discussed in [**?**].*

Homology representatives and paths, not nodes

The heterogeneity above is clarified by recalling that a persistent H_1 feature is not identified with any single concept or set of concepts, but with the equivalence class of cycles that carry the homology class. Under the Vietoris–Rips construction at filtration scale ε, the "ring" of the bee agent is represented by any 1-cycle in the complex $\mathrm{VR}_\varepsilon(\Sigma, d_{\mathcal{A}})$ that is non-trivial in H_1. When NECTAR is overwritten, the cycle

$$\text{FLOWER} \to \text{NECTAR}_{\text{orig}} \to \text{CARRY} \to \text{HIVE} \to \text{DANCE} \to \text{FLY} \to \text{FLOWER}$$

ceases to exist at low filtration scale, but a homologous cycle

$$\text{FLOWER} \to \text{FOOD} \to \text{CARRY} \to \text{HIVE} \to \text{DANCE} \to \text{FLY} \to \text{FLOWER}$$

exists at only a slightly larger scale. The H_1 feature survives the overwrite because the homology representative migrates to an available alternative path. When any other loop concept is overwritten, no alternative path exists, and the feature dies.

The appropriate structural description of a frame is therefore the equivalence class of cycles representing its H_1 homology, not the set of concepts that happen to lie on the canonical representative. Overwrite destabilizes frames by removing vertices from paths; it destroys frames by removing vertices whose removal disconnects all representatives.

A closed-form relationship between cycle geometry and lifetime

The second set of experiments quantifies the relationship between the cycle's geometric configuration and the resulting H_1 lifetime. We sweep NECTAR's angular position within the ring subspace from 0 to π (leaving all other concepts fixed) and record the dominant H_1 lifetime at each angle. Radial shrinkage is confirmed to have no effect (scale invariance of angular distance; std $< 10^{-6}$ over twenty scales).

Figure 2 (top-left) shows a plateau structure: the lifetime is at baseline (0.293) for any NECTAR position within a safe sector, and drops sharply to 0.139 outside that sector. The sector width matches the configuration in which NECTAR fills a large gap in the cycle—that is, the configuration in which NECTAR acts as a critical node.

For any concept configuration that preserves a single H_1 cycle and leaves the death scale of that cycle fixed at 1.0 (as the non-loop landscape in the bee construction does), we obtain the following closed-form relationship.

Proposition 4 (Lifetime-gap relationship; constructed setting)**.** *Let $\mathcal{A}$ be an agent whose concept inventory contains a persistent H_1 feature realized by unit-magnitude centroids in a two-channel subspace, with angular positions forming a set $\Theta \subset [0, 2\pi)$. Let*

$$g(\Theta) = \max_i \left(\theta_{(i+1)} - \theta_{(i)}\right)$$

be the maximal adjacent gap (computed cyclically on the sorted positions). Under the further assumption that the non-loop concepts reside at uniform angular distance 1.0 *from every ring point and therefore trigger H_1 death at filtration scale $\varepsilon = 1.0$, the dominant H_1 lifetime is*

$$L(\mathcal{A}) = 1 - \sqrt{1 - \cos g(\Theta)}.$$

The relationship is stated under the explicit construction above. Its extension to agents whose non-loop landscape departs from the uniform-cone assumption is addressed in the noise-robustness analysis below.

Remark 7 (Scope disclaimer)**.** *This relationship should not be interpreted as a general law of persistence, but as an exact characterization of the constructed regular configuration used for calibration. Its primary role is to provide a sanity-check baseline against which deviations under metric perturbation (noise-robustness subsection below) can be interpreted.*

Figure 2 (bottom-left) compares this analytic prediction to the observed lifetimes across all tested NECTAR angular positions: the prediction matches the observation with mean absolute residual below 10^{-4} (equal to measurement floating-point precision).

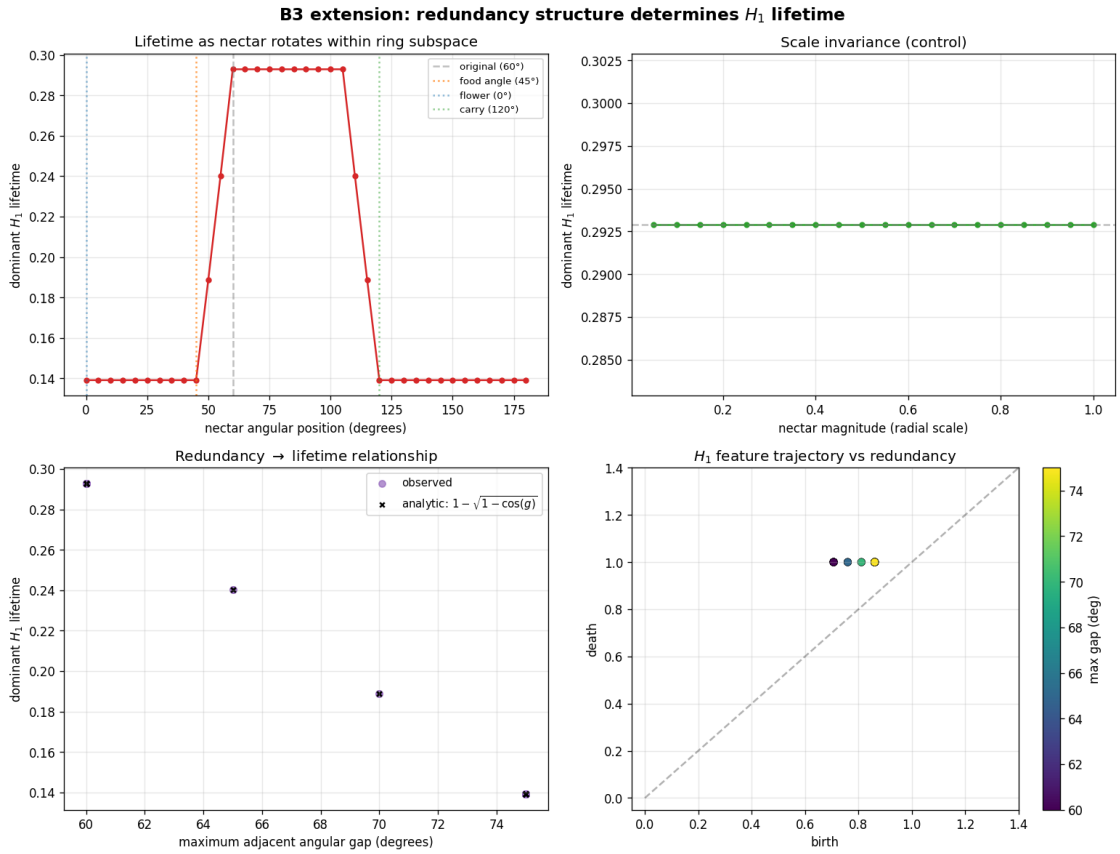

Figure 2: Redundancy structure determines H_1 lifetime. Top-left: lifetime as NECTAR rotates within the ring subspace; plateau shape reflects the safe angular sector bounded by FOOD at 45° and CARRY at 120°. Top-right: radial shrinkage control; scale invariance confirmed. Bottom-left: observed lifetimes (purple) and the analytic prediction of Proposition 4 (black crosses); the overlap is exact to floating-point precision. Bottom-right: H_1 feature trajectory in birth-death plane, coloured by the induced maximal adjacent gap.

Robustness to metric noise

A reviewer may reasonably ask whether Proposition 4's exact match is an artefact of the clean geometric construction or a structural fact of the formula. We therefore tested the relation under two independent noise sources: *ring noise*, in which each ring concept's angular position is perturbed by $\theta \to \theta + \eta$, $\eta \sim \mathcal{N}(0, \sigma_R^2)$, and *non-loop noise*, in which the non-loop concept centroids receive Gaussian perturbations of magnitude σ_{NL} in all six channels (breaking the uniform orthogonal-cone assumption that fixes the death scale at $\varepsilon = 1.0$).

Fifty trials were run for each $(\sigma_R, \sigma_{\mathrm{NL}})$ pair on a 3×4 grid with $\sigma_R \in \{0, 0.03, 0.10\}$ and $\sigma_{\mathrm{NL}} \in \{0, 0.02, 0.05, 0.10\}$. The mean absolute residual $|L_{\mathrm{obs}} - L_{\mathrm{pred}}|$ exhibits the following structure:

	$\sigma_{\mathrm{NL}} = 0$	0.02	0.05	0.10
$\sigma_R = 0$	0.000	0.004	0.009	0.018
$\sigma_R = 0.03$	0.000	0.004	0.009	0.017
$\sigma_R = 0.10$	0.000	0.004	0.008	0.018

Two structural facts emerge. First, the residual is essentially invariant in σ_R: ring perturbations up to 0.10 radians (roughly 5.7°) produce no change in the formula's accuracy. This reflects that Proposition 4 depends on the maximal adjacent gap g of the ring, not on the regularity of the ring; as long as the configuration remains a single H_1 cycle with food as its non-loop substitute vertex, arbitrary displacement of ring concepts does not affect the formula's validity. Second, the residual scales linearly with σ_{NL}: non-loop perturbations of magnitude σ_{NL} produce residuals of approximately $0.18 \cdot \sigma_{\mathrm{NL}}$. This is consistent with the Cohen–Steiner stability theorem for persistence diagrams [**?**], which bounds the bottleneck distance between diagrams of σ-close metric spaces by σ; the observed linear scaling is the expected order for this bound.

Both facts support Proposition 4 as a structural relation rather than a numerical artefact: it is exact whenever the qualitative assumptions (single H_1 cycle; uniform non-loop cone) hold, and degrades gracefully and predictably when they do not.

Ring-size generality within the constructed setting

Proposition 4 as stated does not specialize to hexagonal rings. Within the same constructed setting (regular L-gon embedded in a two-channel subspace; non-loop concepts at uniform angular distance 1.0), we computed the dominant H_1 lifetime for $L \in \{3, 4, 5, 6, 7, 8, 9, 10, 12\}$. For $L \geq 5$ the observed lifetime matches $1 - \sqrt{1 - \cos(2\pi/L)}$ to within floating-point precision (residual $< 10^{-4}$). For $L = 3$, the formula returns a negative value (≈ -0.22), consistent with the observation that a triangle in the Vietoris–Rips complex is a 2-simplex and therefore cannot support a persistent H_1 class; for $L = 4$, the formula returns exactly zero, reflecting the boundary case where the ring's birth and death scales coincide.

We state this observation narrowly: within the tested range of regular L-gons under the constructed metric setting, the proposition holds exactly. Extension to irregular rings, to rings outside the unit-magnitude convention, or to agents with heterogeneous non-loop landscapes is not established by the present experiments.

Remark 8 (Implications of Proposition 4)**.** *The proposition gives the effect of a premature overwrite an explicit functional form. An overwrite of concept c at overwrite fraction t modifies the angular position θ_c in a way that determines the new maximal gap $g(\Theta_t)$ and hence the new lifetime $L(\mathcal{A}_t)$. Three regimes emerge:*

- *If the overwrite moves c into an angular region where a non-loop concept already exists, g remains equal to the baseline (redundant substitution); L is unchanged.*
- *If the overwrite leaves c's sector empty but the landscape contains a nearby non-loop concept that can serve as a substitute, g increases by a bounded amount; L degrades smoothly.*
- *If the overwrite leaves c's sector empty with no nearby non-loop concept, g doubles (the c-to-neighbour edge disappears); if the doubled g exceeds the death-inducing scale, L collapses to zero.*

This three-regime structure is the quantitative expression of the qualitative heterogeneity observed in the first experiment set.

Falsifiable prediction for empirical studies

The closed-form relationship of Proposition 4 supplies a falsifiable prediction for empirical studies of fine-tuning under the persistence-signature protocol. For a model whose current interpretive frame is characterized by a persistent H_1 feature with a known homology representative, perturbations targeting concepts on critical paths of that representative should produce lifetime reductions following the gap formula; perturbations targeting redundantly-placed concepts should produce negligible lifetime reduction. The test requires (i) identifying the current dominant frame's H_1 representative via mechanistic interpretability methods, (ii) classifying concepts as critical or redundant by structural role within the representative, and (iii) measuring post-fine-tuning lifetime change. The structural classification can be obtained from the Vietoris–Rips complex alone, without assuming any semantic interpretation of which concepts are "important."

Reversibility in the static toy

A reversibility check was performed: the bee centroids were perturbed to $t = 1$ and then restored to $t = 0$. The H_1 lifetime recovered exactly to the baseline value of 0.293. This exact reversibility is a feature of the static toy model—the experiment measures only the instantaneous persistence signature at each perturbation state and does not model the downstream inferences that the overwrite would trigger in a dynamically operating agent. The non-reducibility claim of Proposition 2 concerns the latter; operationalizing it requires a dynamic experimental design in which downstream inferences are cached and subject to their own persistence measurement, which we identify as a subsequent research target.

Summary

The persistence-signature framework renders the overwrite dynamics of Section **??** empirically tractable at the level of static persistence. The four observations of the experiment set can be summarized as follows:

1. Continuous overwrite produces graded, non-linear degradation of the frame's H_1 lifetime, with threshold behaviour near $t \approx 0.5$ in the present construction.
2. Frame constituents are structurally heterogeneous: concepts with nearby non-loop substitutes are redundantly placed, while concepts without substitutes are critical nodes.
3. The homology representative of a surviving H_1 feature migrates to the alternative path provided by the substitute; the feature itself is an equivalence class of cycles, not a set of vertices.
4. The dominant H_1 lifetime is an analytic function of the induced maximal adjacent gap in the cycle (Proposition 4); this gives overwrite bias propagation a closed-form quantitative expression.

Observation 2 reformulates the standard "all overwrites propagate bias" claim into a structurally graded statement whose testable content is stronger than the original. Observations 3 and 4 provide the topological substrate for that reformulation.

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